

W2-PM-FINAL

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

CYBERSECURITY | NETWORKWALKS

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Program / Batch	B083D – Networkwalks Cybersecurity Internship
Date	15 September 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client / Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network (Wi-Fi hotspot)
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance and scanning activities
WHOIS	Find domain registration details (owner, dates, name servers)
whatweb	Fingerprint web technologies (server, CMS, plugins, IP)
nslookup	Resolve the domain name to its IP address using DNS
curl -I	Read the HTTP response headers of the website
wafw00f	Detect whether a Web Application Firewall protects the site
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV)
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses
Windows CMD	Local IP and MAC address identification (ipconfig)

4. Activities Performed

4.1 Footprinting & Reconnaissance (W2-PM1)

I performed reconnaissance against the **networkwalks.com** domain using six Kali Linux tools: **WHOIS**, **whatweb**, **nslookup**, **curl**, **wafw00f** and **dnsrecon**. Each tool was used to collect a different type of information about the target.

Task 1 – WHOIS

Command used:

```
whois networkwalks.com
```

Key Findings: Domain registered with GoDaddy.com, LLC. Creation date 2019-11-06, Expiry 2027-11-06. Name servers point to HostGator (NS6135.HOSTGATOR.COM / NS6136.HOSTGATOR.COM). DNSSEC is unsigned. Abuse contact: abuse@godaddy.com.

```

kali@kali: ~
$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452219235_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T18:08:14Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp/clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp/clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp/clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp/clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2025-09-15T15:41:54Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this data only
for lawful purposes and that under no circumstances will you use this data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right

```

Figure 1: WHOIS output for networkwalks.com

How attackers use this: WHOIS reveals the registrar, registration & expiry dates, and name servers. The name servers point to HostGator, so an attacker instantly learns the hosting provider.

Task 2 – whatweb

Command used:

```
whatweb networkwalks.com
```

Key Findings: Apache web server, WordPress 7.1, WordPress Download Manager 3.3.58, jQuery 3.7.1, Bootstrap 7.1, real IP 192.232.216.135, email info@networkwalks.com, title “Networkwalks Academy”.

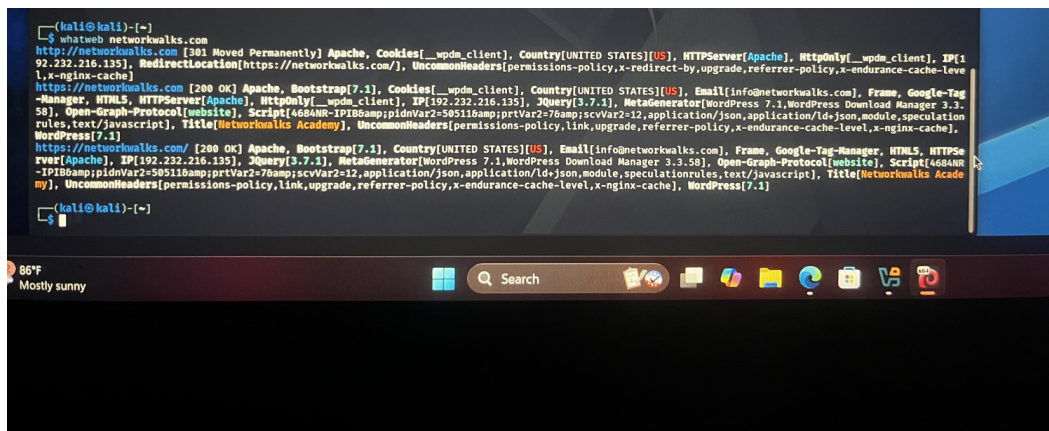


Figure 2: whatweb fingerprinting results

How attackers use this: whatweb exposes exact software versions. An attacker searches vulnerability databases for known exploits against WordPress 7.1 and the detected plugin.

Task 3 – nslookup

Command used:

```
nslookup networkwalks.com
```

Key Findings: Domain resolves to **192.232.216.135** (non-authoritative answer).

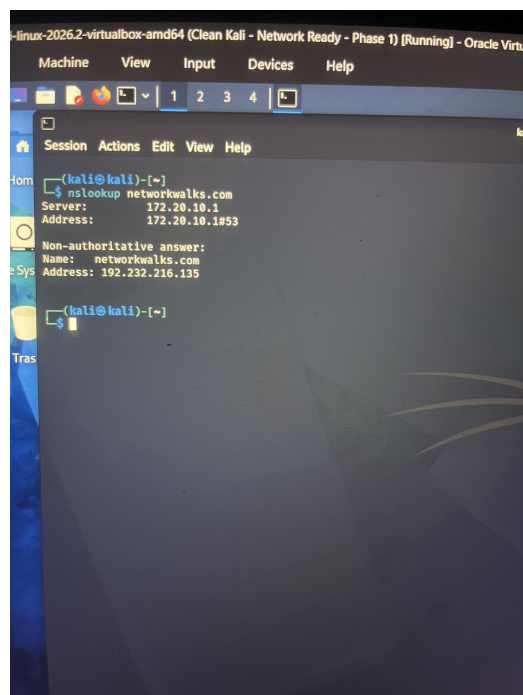


Figure 3: nslookup resolution

How attackers use this: Knowing the real IP lets an attacker scan the server directly and map infrastructure.

Task 4 – curl -I

Command used:

```
curl -I https://networkwalks.com
```

Key Findings: HTTP/2 200, Server: Apache, exposes WordPress REST API endpoint /wp-json/, sets __wpdm_client cookie, x-nginx-cache: WordPress.

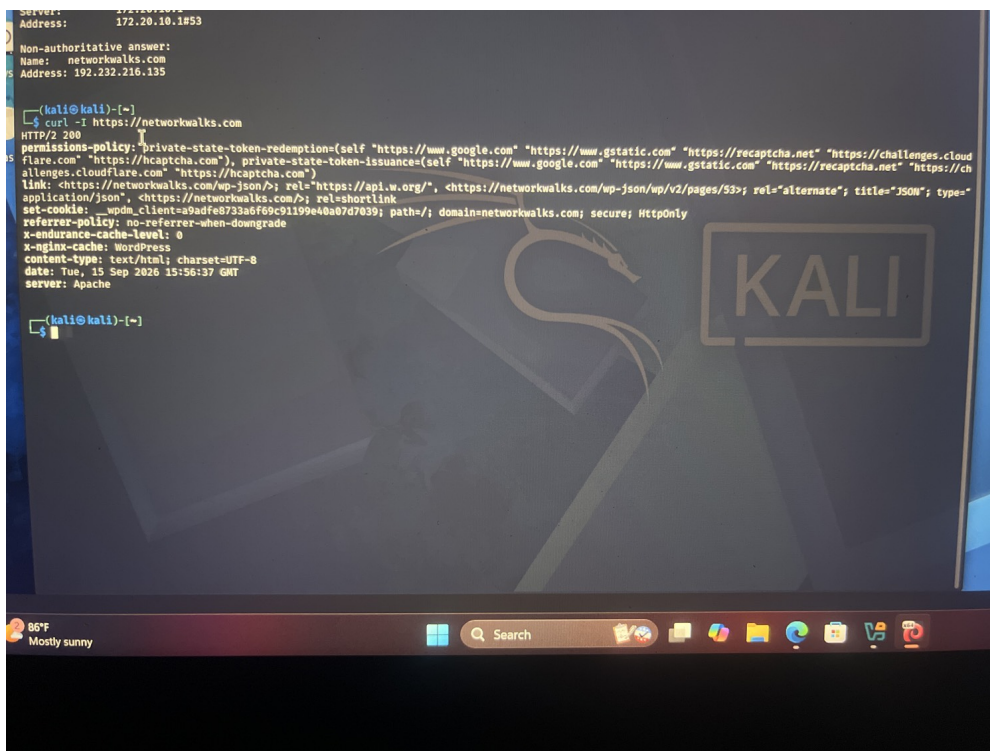


Figure 4: HTTP response headers

How attackers use this: The /wp-json/ endpoint is a valuable discovery surface for further enumeration without loading the full page.

Task 5 – wafw00f

Command used:

```
wafw00f 192.232.216.135
```

Key Findings: The site is behind ModSecurity (SpiderLabs) WAF.

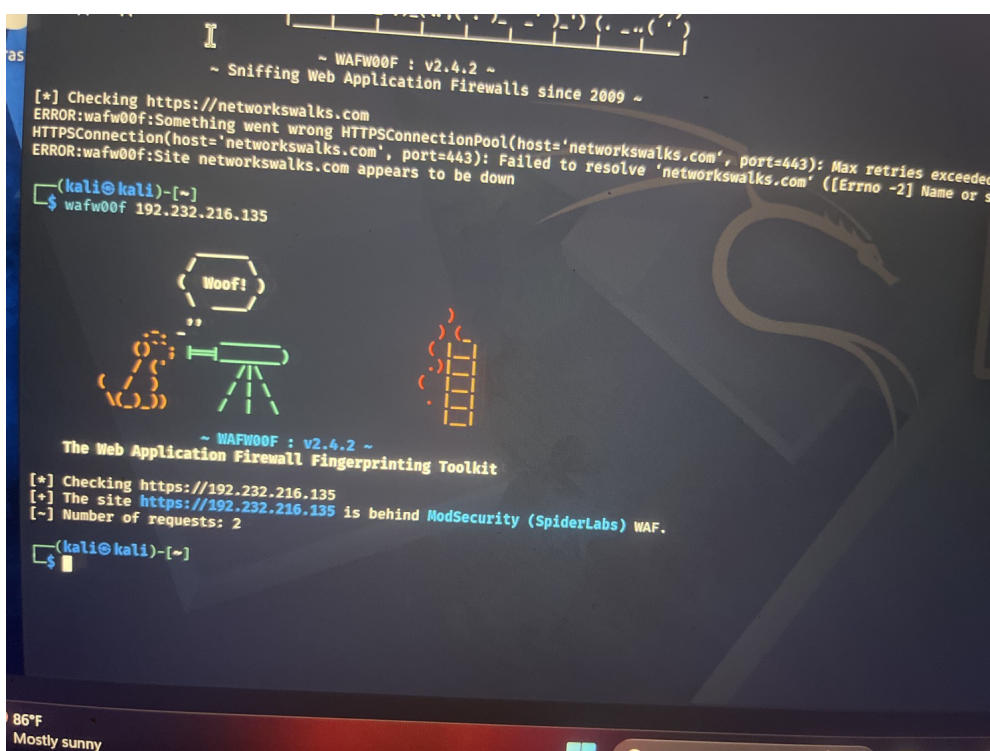


Figure 5: wafw00f detection result

How attackers use this: Knowing a WAF is present shapes the whole attack – naive attempts will be blocked or logged, so the attacker must adapt or attempt bypass techniques.

Task 6 – dnsrecon

Command used:

```
dnsrecon -d networkwalks.com
```

Key Findings: NS records point to HostGator (Bind 9.16.23-RH), MX mail.networkwalks.com → 192.232.216.135, SPF record present, multiple cPanel autodiscover SRV records.

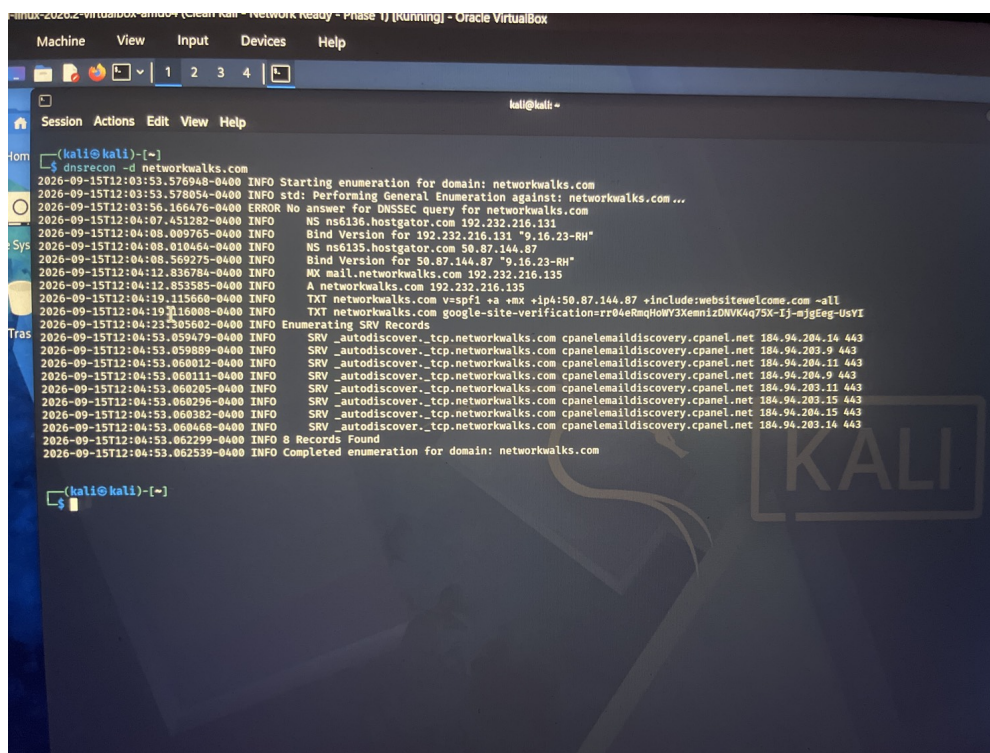


Figure 6: dnsrecon full enumeration

How attackers use this: dnsrecon maps the entire DNS footprint – mail servers, DNS version, SPF policy and cPanel service records. Each record is a potential foothold.

4.2 Network Scanning with Zenmap (W2-PM5)

For the second activity I used **Zenmap** to perform network discovery on my own local Wi-Fi network (phone hotspot). The practical required identifying my local IP address and subnet, discovering live hosts, collecting IP and MAC addresses, and generating a network topology.

Local Network Configuration (ipconfig)

My active Wi-Fi adapter showed:

- IPv4 Address: **172.20.10.3**
- Subnet Mask: **255.255.255.240** → Subnet **172.20.10.0/28**
- Default Gateway: 172.20.10.1

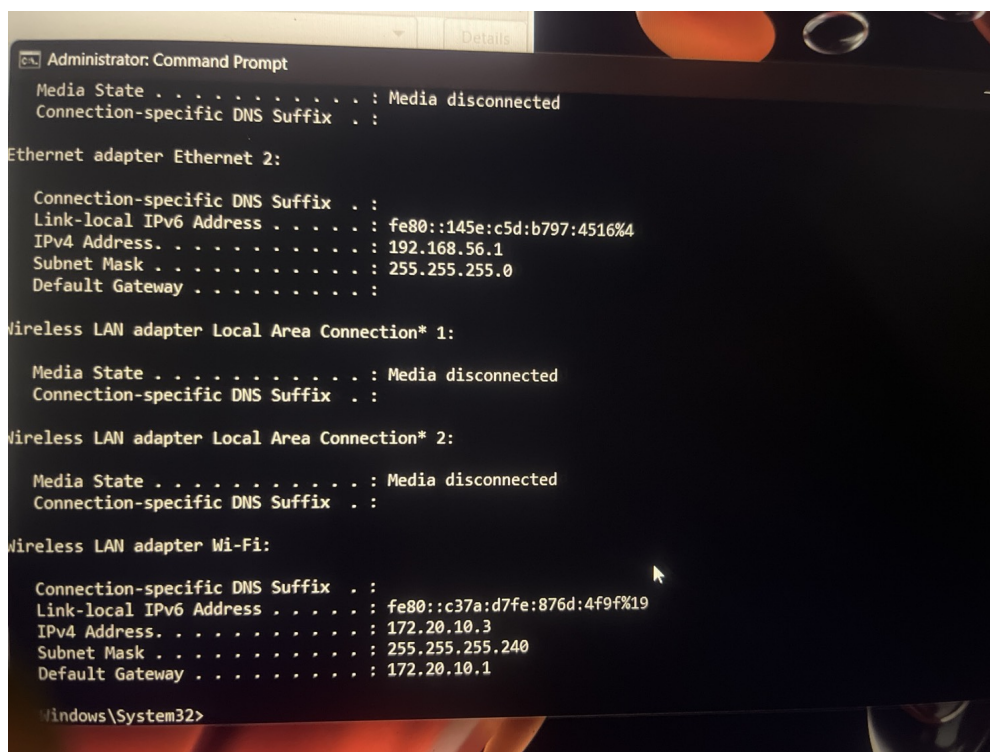


Figure 7: ipconfig – local IP and subnet

Ping Scan Results

Command / Profile used in Zenmap: Target = 172.20.10.0/28 , Profile = Ping scan

Live hosts found: 2

- 172.20.10.1 – Gateway (phone hotspot) – MAC: 2A:02:44:3F:08:64
- 172.20.10.3 – My own Windows PC

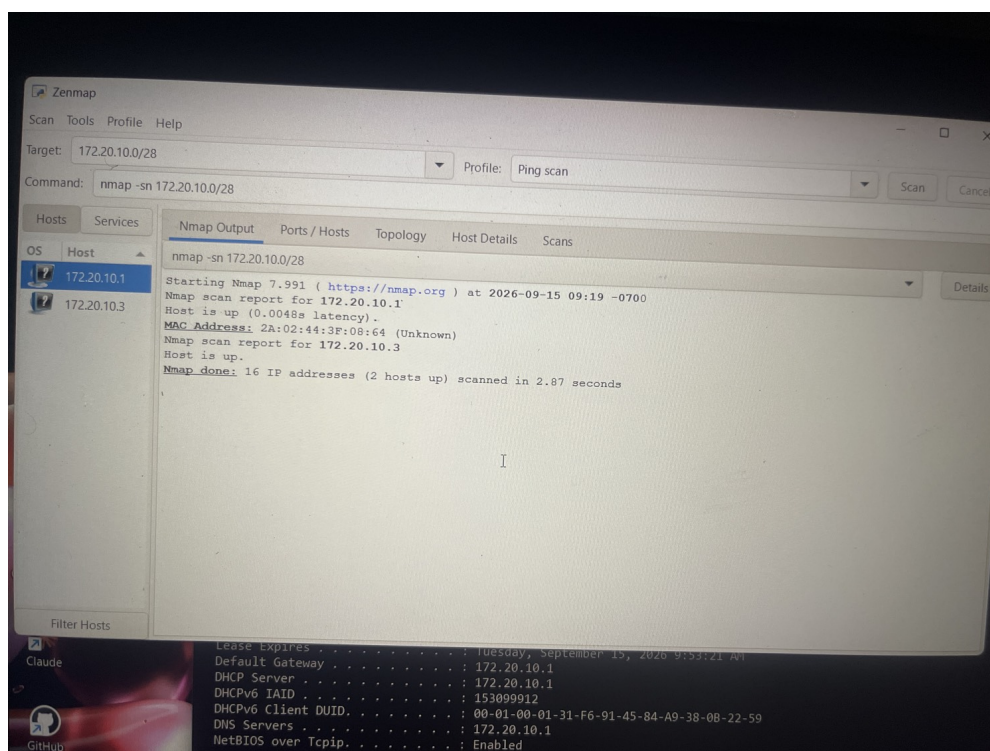


Figure 8: Zenmap Ping Scan results

Network Topology

After the scan I opened the Topology tab, enabled the legend, and saved the graphic as PDF. The topology clearly shows the two live hosts connected.

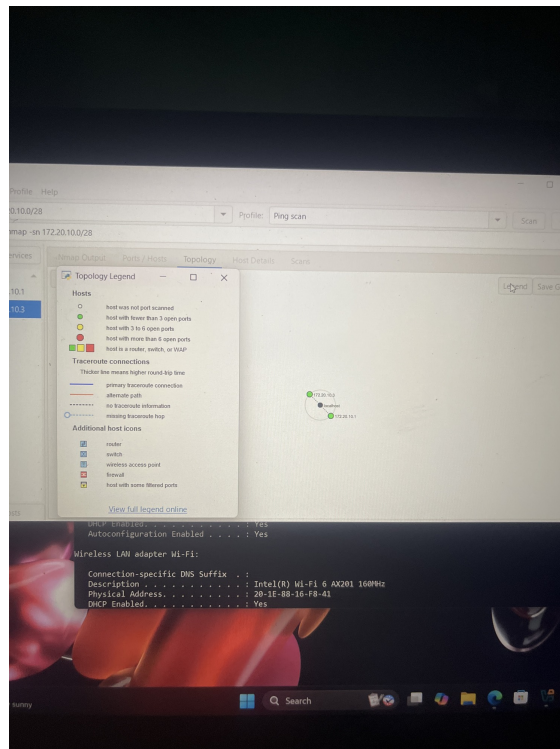


Figure 9: Network Topology (2 live hosts)

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence	Potential Impact	Level
1	Web technology information exposed	whatweb: WordPress 7.1 + WP Download Manager 3.3.58	Attackers may search for known exploits against these versions	Medium
2	Server IP address identifiable	nslookup → 192.232.216.135	Enables direct scanning of the server	Low
3	HTTP technical information exposed	curl -I shows /wp-json/ REST API	Assists further technology fingerprinting and enumeration	Low
4	WAF technology identifiable	wafw00f: ModSecurity (SpiderLabs)	Reveals security architecture; attacker can plan bypasses	Low
5	DNS infrastructure information exposed	dnsrecon: NS, MX, SPF, cPanel SRV records	Helps build a broader infrastructure profile	Medium
6	Multiple live hosts on local network	Zenmap: 2 hosts on 172.20.10.0/28	Unknown devices could be present on a larger network	Medium

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities. No exploitation or vulnerability validation was performed. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

- 1. Review publicly exposed technology information** – Regularly review what information about web technologies, CMS and plugins is publicly visible.

- 2. Keep software updated** – CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
- 3. Review HTTP headers** – HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
- 4. Review DNS records regularly** – DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
- 5. Properly configure and monitor the WAF** – Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naïve attacks.
- 6. Perform regular internal network discovery** – Organizations should periodically scan their own networks to identify active devices.
- 7. Investigate unknown devices** – Any unexpected device discovered during network scanning should be investigated and verified.
- 8. Maintain network documentation** – Network topology and device information should be documented and updated regularly.
- 9. Perform security testing only with authorization** – Reconnaissance and scanning must only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS provides domain registration data, whatweb identifies web technologies, nslookup resolves domain names, curl inspects HTTP headers, wafw00f detects a WAF, and dnsrecon enumerates DNS records.

In the network scanning activity I used Zenmap to identify my local network configuration (172.20.10.0/28) and discover the two active hosts. I also collected IP and MAC address information and created a network topology.

These exercises demonstrated that information gathering is a critical part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses. All activities were performed strictly within the authorized scope of the internship.

8. Evidences Collected

All screenshots used in this report are stored in the **evidences/** folder of the accompanying GitHub repository:

- 1_whois.jpg – WHOIS domain registration output
- 2_whatweb.jpg – whatweb technology fingerprint
- 3_nslookup.jpg – DNS resolution result
- 4_curl.jpg – HTTP response headers
- 5_wafw00f.jpg – WAF detection (ModSecurity)
- 6_dnsrecon.jpg – Full DNS enumeration
- 7_ipconfig.jpg – Local IP & subnet identification
- 8_zenmap_scan.jpg – Zenmap Ping Scan results
- 9_topology.jpg – Network Topology graphic

— End of Report —

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Project Information

Program: Cybersecurity program at Networkwalks | Week: 02 | Repository: GitHub